HOMEWORK 2: BRAIN CONDITIONING STATS □

DUE: *Mar 5, 2026 @ 11:59 PM*

24-HR LATE DUE DATE WITH A 15% PENALTY: *Mar 6, 2026 @ 11:59 PM*

Objective:

The aim of this assignment is to deepen students' understanding of statistics and hypothesis testing using Python. By engaging with some theoretical questions as well as practical exercises, students will apply statistical methods and perform hypothesis tests, using Python to code and execute these techniques. This approach will help solidify their grasp of statistical principles and their application in Python, bridging theoretical knowledge with practical skills.

Overview:

- This assignment is worth 120 points.
- There are 3 mandatory parts to complete in this section (DON'T REMOVE ANY PART OF THE QUESTIONS).
- Make a local copy, read all of the instructions carefully, and complete all tasks.

Submission:

- Submit your completed notebook (.ipynb) file and PDF to the "HW2 - Brain Conditioning Stats" section in Gradescope.

Reminder: Please make sure your code runs before submitting your work. Code sections that do not run will receive 0 credits, no partials will be given. This is VERY important in real project development.

For Math problems, check out homework submission requirements and show work accordingly. No credits will be given without appropriate progress.

DO NOT REMOVE ANY PART OF ANY OF THE QUESTIONS OR YOU LOSE CREDIT

No Hardcoding either 😋

Part 1: Statistics Problem Solving

##Q1) (10 POINTS) Bayes Theorem

Suppose some hacker found a dataset on uselessdatasets.com containing information about three different types of users on an online platform: "bloggers", "shoppers", and "reviewers". The data has 13,000 users. There are 5,500 bloggers, 7,000 shoppers, and 6,500 reviewers. The users could be in multiple categories. 2,200 of the bloggers are shoppers, 1,800 of the bloggers are reviewers, and 3,000 shoppers are also reviewers.

Answer the following questions in the designated boxes:

1. (3 POINTS) If X is a random variable that represents the users that were cross listed into all 3 categories, what is the value of X? (Hint: think of a Venn Diagram.)

   13000 = 5500 + 7000 + 6500 − 2200 − 1800 − 3000 + X

Final Answer: X = 1000

1. (3 POINTS) Calculate the probability that a randomly selected shopper is also a blogger. Round to the nearest hundredth. (Hint: Use Bayes Theorem)

   P(B|S) = (B intersect S)/S = 2200/7000 = 0.314...

Final Answer: X = 0.31

1. (4 POINTS) Calculate the probability that a random user is in exactly two categories but not all three. Round to the nearest hundredth.

   (|B intersect S| − X) + (|B intersect R| − X) + (|S intersect R| − X) (2200 - 1000) + (1800 - 1000) + (3000 - 1000) = 1200 + 800 + 2000 = 4000 4000 / 13000 = 0.0307...

Final Answer: 0.31

##Q2) (6 POINTS) Expected Values

A fair six-sided die has outcomes 1, 2, 3, 4, 5, 6, each with probability 1/6.

- Let

$$T=\{(r_1,r_2):r_1,r_2\in\{1,2,3,4,5,6\}\}$$

  Then $|T|=36$.
- Let

$$S=\{(r_1,r_2,r_3):r_1,r_2,r_3\in\{1,2,3,4,5,6\}\}$$

  Then $|S|=216$.

Define X_n to be the sum of the n rolls:

$$X_n=r_1+r_2+\cdots+r_n.$$

In particular:

- For $(r_1,r_2)\in T$:

$$X_2=r_1+r_2.$$

- For $(r_1,r_2,r_3)\in S$:

$$X_3=r_1+r_2+r_3.$$

1. (3 POINTS) What is $E[X_2]$?

Hint: For each possible value of x, think about how many combinations can produce that same value. Group those outcomes together and form a summation over all possible x. Then apply the definition of expectation to compute the result.

Be systematic when counting — organizing the cases carefully will make the pattern much clearer.

Total Combinations: 6 x 6 = 36

How many ways each sum occurs: 2(1) + 3(2) + … + (7)6 + … + (11)2 + (12)1 = 252

252/36 = 7

Final Answer: 7

1. (3 POINTS) What is $E[X_3]$?

Hint: Similar to the previous problem, but now there are 6 times more possible combinations due to one additional roll. The pattern can become quite long, so be careful and systematic when deriving it.

For those with limited statistical background, if you strongly prefer, you may use coding to help solve the problem. However, it is still mandatory to clearly show your reasoning and intermediate progress.

This exception only applies to $E[X_3]$, not $E[X_2]$.

E[X3] = E[D1 + D2 + D3]

E[D] = (1 + 2 + 3 + 4 + 5 + 6)/6 = 3.5

E[X3] = 3(3.5) = 10.5

Final Answer: 10.5

##Q3) (6 POINTS) Probability distribution

Let X be a continuous random variable that follows a normal distribution with mean $\mu = 10$ and standard deviation $\sigma = 2$.

Answer the following questions in the designated boxes:

1. (3 POINTS) What is the probability that X takes a value between 6 and 12? Hints: You may have to utilize the standard normal table: https://math.arizona.edu/~jwatkins/normal-table.pdf

How to read the "Standard Normal Cumulative Probability Table" table:

- Rows and Columns: The rows correspond to the first digit and first decimal place of z. The columns correspond to the second decimal place of z.
- Check out: https://byjus.com/maths/z-score-table/

for 6: z = (6 - 10)/2 = -2

for 12: z = (12 - 10)/2 = 1

P(Z < 1) = 0.8413

P(Z < -2) = 0.22

P(-2 < Z < 1) = 0.8412 - 0.0228 = 0.8185

Final Answer: 0.82

1. (3 POINTS) What is the probability that X takes a value greater than 15?

z = (15 - 10)/2 = 2.5

P(Z < 2.5) = 0.9938

P(Z > 2.5) = 1 − 0.9938 = 0.0062

Final Answer: 0.01

Part 2: Python Warmups

##Q1) (10 POINTS) Bernoulli Trials

Consider a sequence of n Bernoulli trials with success probability p per trial. A string of consecutive successes is known as a *streak*.

> **Task to do:** Write a function that returns a `collections.Counter` dictionary object that maps the length of a streak k to the number of times it is observed in an input sequence `xs`. For example, if `xs = [0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 1]`, the output would be `Counter({1: 2, 2: 1, 3: 2})`. We have imported `Counter` from the Python `collections` library for you in the code block below. The order of the keys in the Counter does not matter, unsorted is fine.

```python
from collections import Counter

def count_streaks(xs):
    """Count number of success runs of length k."""

    streak_lengths = [] # store the length of every streak we observe
    current_streak = 0 # keep track of current streak length

    for x in xs:
        # if we see a 1, increment the current streak length
        if x == 1:
            current_streak += 1
        # if we see a 0, end the current streak and store its length
        else:
            if current_streak > 0:
                # save completed streak length
                streak_lengths.append(current_streak)
                # reset counter for next streak
                current_streak = 0

    # sequence ends with a streak
    if current_streak > 0:
        streak_lengths.append(current_streak)

    xs = streak_lengths

    return Counter(xs)

# Use this cell to test your answer. MAKE SURE YOUR RESULTS ARE SHOWN
BELOW AFTER RUNNING THIS BOX
import numpy as np
print(count_streaks([0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 1]))
np.random.seed(0)
count_streaks(np.random.randint(0,2,1000000))
```

```
Counter({1: 2, 3: 2, 2: 1})
```

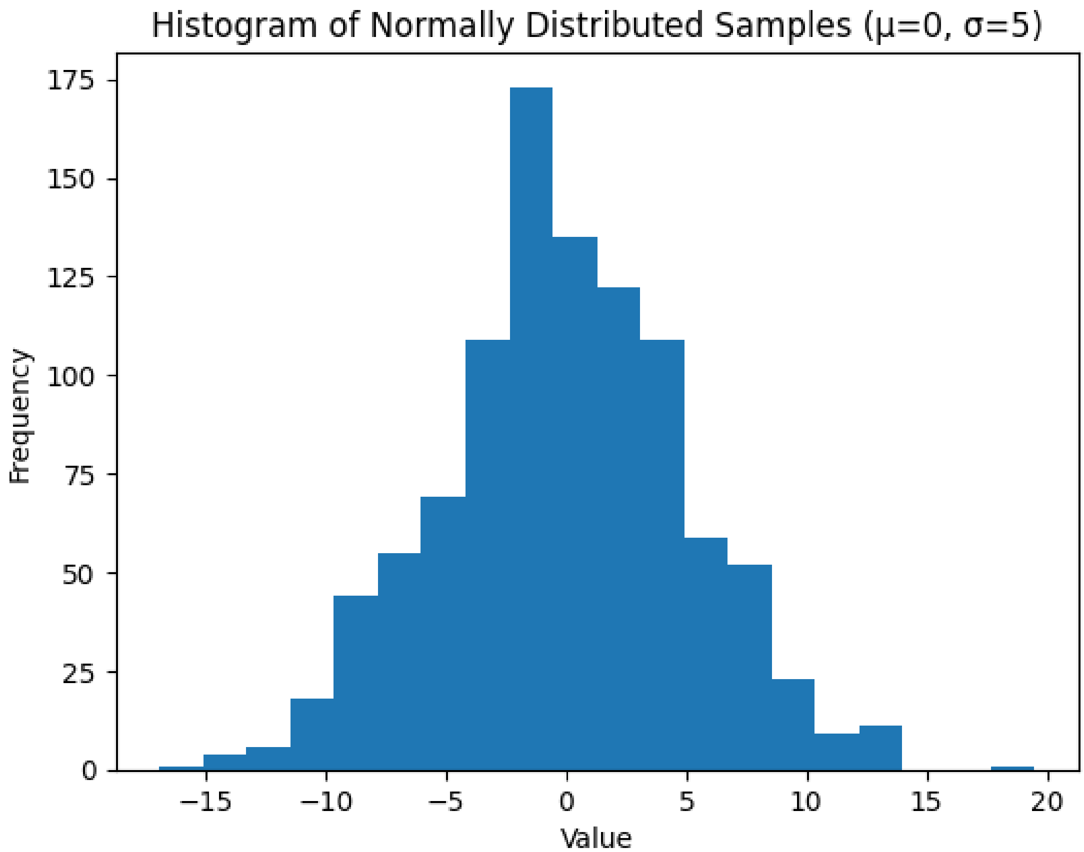
```
Counter({1: 125036,
         2: 62589,
         3: 31100,
         4: 15859,
         5: 7699,
         6: 3893,
         7: 1921,
         8: 946,
         9: 470,
         10: 245,
         11: 126,
         12: 45,
         13: 29,
         14: 11,
         15: 9,
         17: 6,
         16: 2,
         18: 1})
```

##Q2) (10 POINTS) Distribution and Visualization

The goal of solving this problem is to become familiar with using built-in Python libraries to create various distributions. Plotting serves as an initial step toward data visualization.

1. (3 POINTS) Create a normally distributed random variable with mean $\mu = 0$, standard deviation $\sigma = 5$ and sample size $n = 1000$. Plot the histogram. Add labels and titles and other details as desired to make your plot understandable. You must use the packages `numpy` and `matplotlib`.

```
import numpy as np
import matplotlib.pyplot as plt

mu = 0       # mean
sigma = 5    # sd
size = 1000 # num samples

# generate 1000 random values from a normal distribution with mean 0
and sd 5
samples = np.random.normal(mu, sigma, size)

plt.hist(samples, bins=20)

plt.xlabel("Value")
plt.ylabel("Frequency")
plt.title("Histogram of Normally Distributed Samples (μ=0, σ=5)")

plt.show()
```



1. (7 POINTS) We are exploring the Central Limit Theorem (CLT) using a Poisson distribution. Suppose you have a population that follows a Poisson distribution with a rate parameter (or mean) $\lambda = 3$. You will draw multiple samples from this population and calculate the mean of each sample.

Write a Python function that simulates this process. The input of the function should be the sample size, the number of samples, and lambda. The function should:

1. Generate a population with a Poisson distribution (check: https://numpy.org/doc/stable/reference/random/generated/numpy.random.poisson.html).
2. Draw multiple samples and calculate the mean of each sample.
3. Return these means as an iterable.

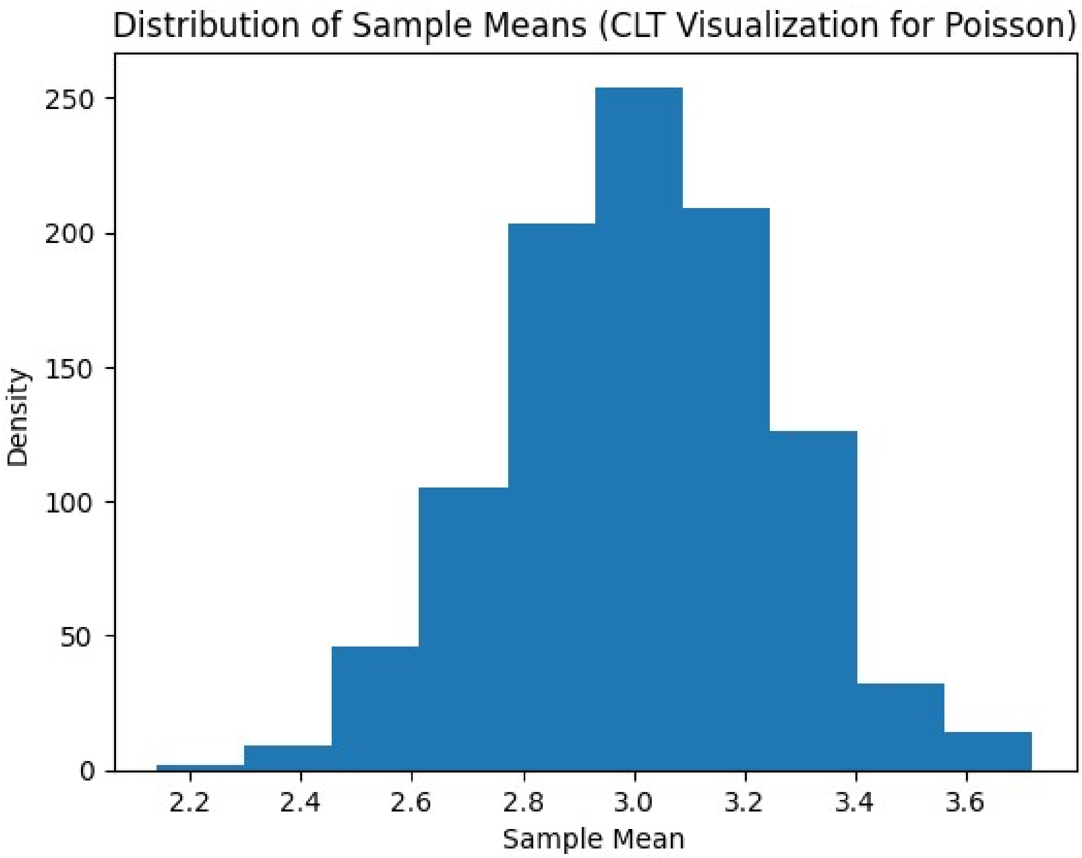
There will be no partial credit granted for this question. Any hardcoded results will receive a 0.

```
import numpy as np

def poisson_clt_simulator(sample_size, num_samples, lambda_):
    sample_means = [] # store the mean of each sample
    for _ in range(num_samples):
        # generate one sample from a poisson distribution with mean 
lambda_
```

```
        sample = np.random.poisson(lam=lambda_, size=sample_size)
        sample_means.append(np.mean(sample)) # compute the mean of 
this sample and store it
    return sample_means
```

Now use the function to generate 1,000 sample means with sample size 50. Plot the distribution of these sample means to visualize the Central Limit Theorem. Add labels and titles and other details as desired to make your plot understandable.

```
import matplotlib.pyplot as plt

# parameters
sample_size = 50
num_samples = 1000
lambda_ = 3

# simulate sample means from poisson distribution
sample_means = poisson_clt_simulator(sample_size, num_samples, 
lambda_)

plt.hist(sample_means)

plt.xlabel('Sample Mean')
plt.ylabel('Density')
plt.title('Distribution of Sample Means (CLT Visualization for 
Poisson)')

plt.show()
```



##Q3) (18 POINTS) More on Distributions

You can't get around with distributions while data sciencing. Let's explore how distributions are related to each other.

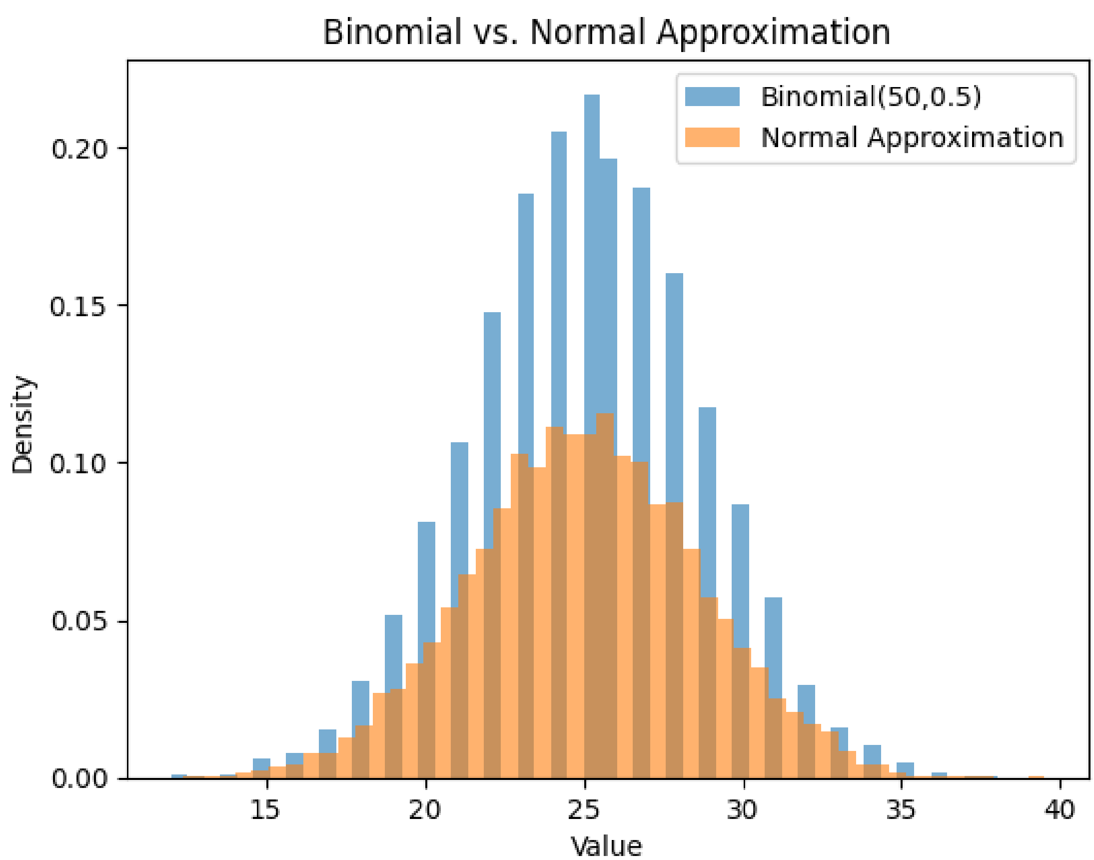
1. (6 POINTS) Since we have successfully demonstrated how CLT works, lets see what we can do with it.

Check out https://numpy.org/doc/stable/reference/random/generated/numpy.random.binomial.html for how to create independent binomial distributions

TASK: Show that a Binomial(n, p) distribution approximates a Normal distribution when n is LARGE (due to CLT). Complete the following code according to comments.

```
import numpy as np
import matplotlib.pyplot as plt

size = 10000
n, p = 50, 0.5  # large n for normal approximation
binomial_samples =  np.random.binomial(n=n, p=p, size=size)
normal_samples = np.random.normal(loc=n*p, scale=np.sqrt(n*p*(1-p)),
size=size) # Don't worry about this line unless you are interested
```

```
plt.hist(binomial_samples, bins=50, density=True, alpha=0.6,
label="Binomial(50,0.5)")
plt.hist(normal_samples, bins=50, density=True, alpha=0.6,
label="Normal Approximation")
plt.legend()
plt.title("Binomial vs. Normal Approximation")
plt.xlabel("Value")
plt.ylabel("Density")
plt.show()
```



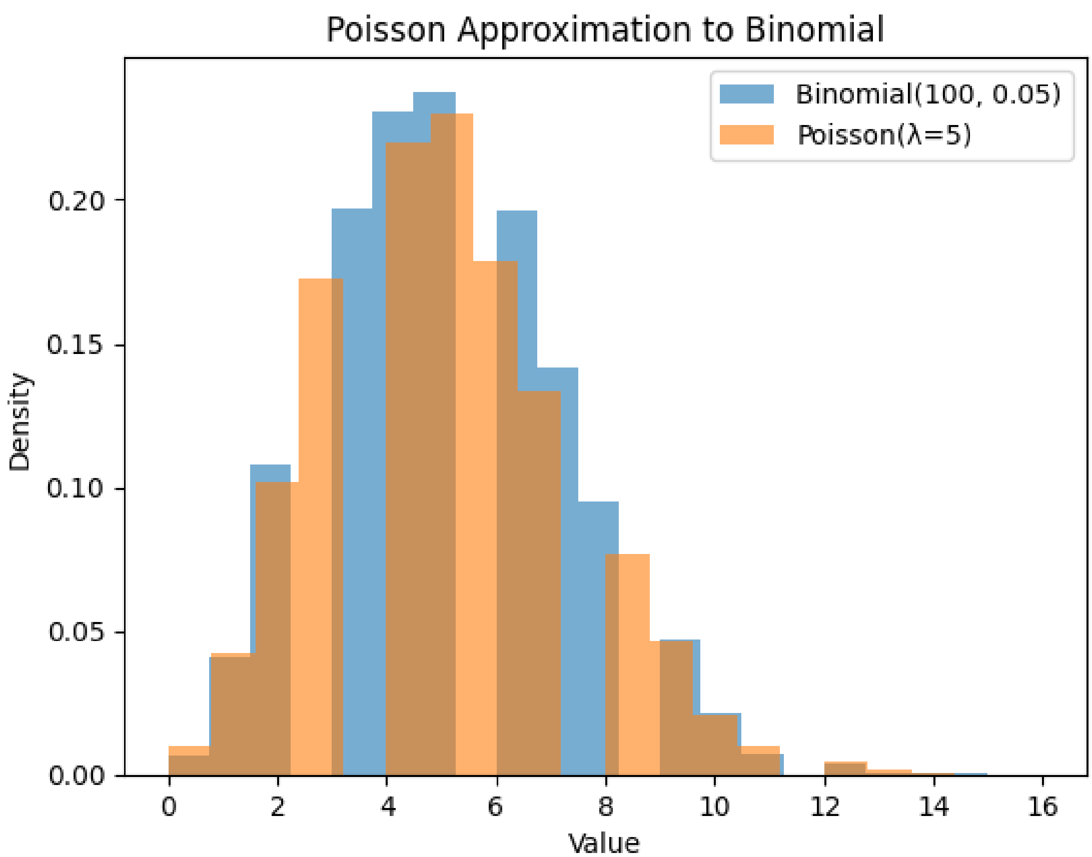
2. (6 POINTS) Now with Poisson

Check out https://numpy.org/doc/stable/reference/random/generated/numpy.random.poisson.html for how to create independent poisson distributions

TASK: Show that when n is large and p is small, a Binomial(n, p) distribution approximates a Poisson distribution with $\lambda = np$. Complete the following code according to comments.

```
size = 10000
n, p = 100, 0.05  # np = 5, small p
binomial_samples = np.random.binomial(n=n, p=p, size=size)
poisson_samples = np.random.poisson(lam=n*p, size=size)
```

```
plt.hist(binomial_samples, bins=20, density=True, alpha=0.6,
label="Binomial(100, 0.05)")
plt.hist(poisson_samples, bins=20, density=True, alpha=0.6,
label="Poisson(λ=5)")
plt.legend()
plt.title("Poisson Approximation to Binomial")
plt.xlabel("Value")
plt.ylabel("Density")
plt.show()
```



3. (6 POINTS) Poisson and Exponential

We know that Poisson counts the number of arrivals, while Exponential models the time between them.

TASK: Generate a Poisson distribution and an Exponential distribution for plotting. You do not have to describe and justify your findings, just witness how they are correlated.

Check out
https://numpy.org/doc/stable/reference/random/generated/numpy.random.exponential.html

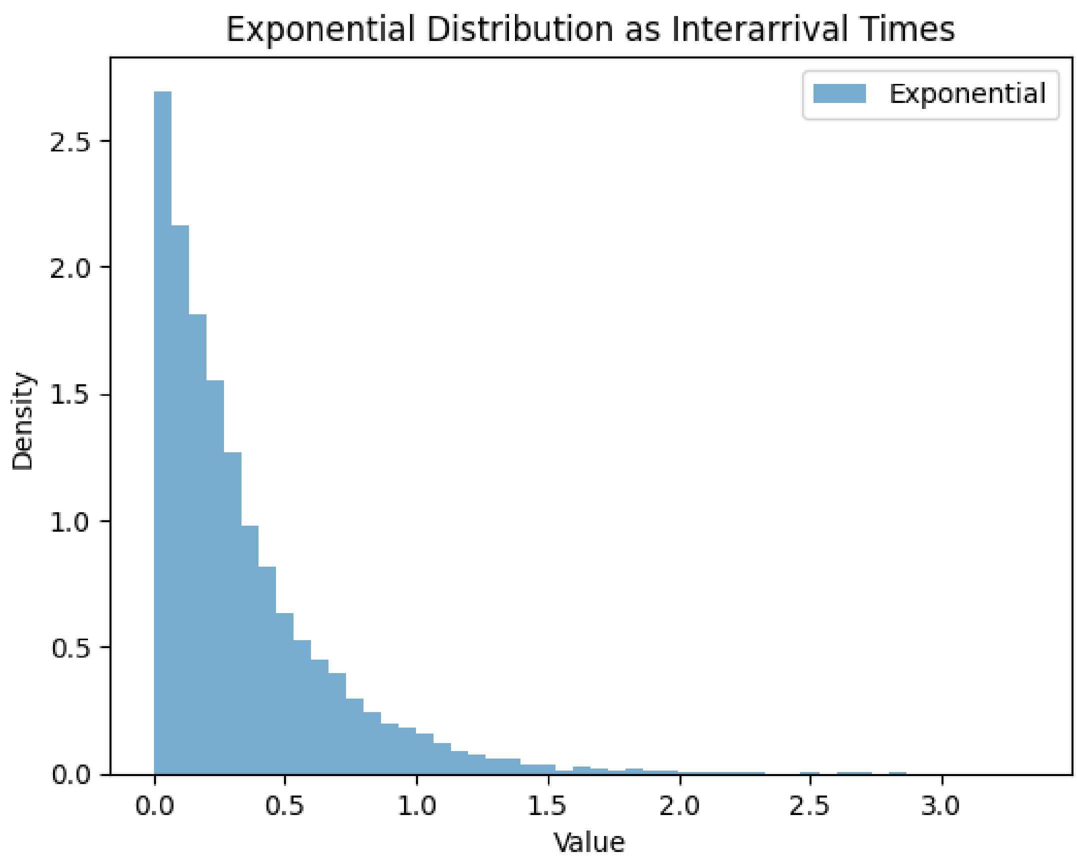
NOTES: If you dont know about Exponensial Distribution, check out:

- https://www.probabilitycourse.com/chapter4/4_2_2_exponential.php

- https://www.ncl.ac.uk/webtemplate/ask-assets/external/maths-resources/business/probability/exponential-distribution.html

Complete the following code according to comments.

```
size = 10000
lambda_exp = 3  # rate for Poisson

poisson_time_intervals = np.random.poisson(lambda_exp, size)
exponential_samples = np.random.exponential(scale=1/lambda_exp,
size=size)

plt.hist(exponential_samples, bins=50, density=True, alpha=0.6,
label="Exponential")
plt.legend()
plt.title("Exponential Distribution as Interarrival Times")
plt.xlabel("Value")
plt.ylabel("Density")
plt.show()
```



Part 3: Hypothesis Testing

##Q1) (14 POINTS) Hypothesis Tests and P_value

TASK: For the next 5 problems, please describe when you would use each hypothesis test. Answer the questions in the designated boxes:

- Chi-Squared Test
- Z test
- T test
- Mann-Whitney U Test
- Anova

1.1 (2 POINTS) Chi-Squared Test

> Chi-Squared Test would be used when analyzing categorical data to determine if there is a significant association between two nominal variables. It is a good choice for testing whether the frequency distribution of one variable differs across the levels of another.

1.2 (2 POINTS) Z-Test

> Z-Test would be used when comparing a sample mean to a known population mean while knowing the population standard deviation. This test requires a large sample size, typically greater than 30, and assumes the data follows a normal distribution.

1.3 (2 POINTS) T-Test

> T-Test would be used when comparing the means of two groups when the population standard deviation is unknown. It functions as an independent test for two separate groups or as a paired test for measurements taken from the same group at different times.

1.4 (2 POINTS) Mann-Whitney U Test

> Mann-Whitney U Test would be used when comparing two independent groups but the data does not follow a normal distribution. As a non-parametric alternative to the T-test, it focuses on the ranks of the data points rather than the actual means.

1.5 (2 POINTS) ANOVA Test

> ANOVA would be used when comparing the means of three or more independent groups simultaneously. It determines if at least one group mean is significantly different from the others without increasing the risk of a Type I error.

1.6 (4 POINTS) : Explain the statistical interpretation of a p-value. What is a p-value? What does it mean? Be sure to explain beyond just "rejecting or failing to reject the null hypothesis."

> A p-value represents the probability of seeing data as extreme as the current results if the null hypothesis is actually true. Instead of just being a rule for rejecting a hypothesis, this value measures how "weird" or surprising the findings would be in a world where no effect or difference exists. A very low p-value means the outcome is so

unlikely to happen by luck that the null hypothesis starts to look incorrect. On the other hand, a high p-value suggests the data is totally consistent with random chance, meaning there is no strong reason to doubt the original assumption. It is important to note that this number describes the data itself rather than the chance of a theory being right or wrong.

##Q2) (2 POINTS) Create a DataFrame and Display

```python
import pandas as pd
import matplotlib.pyplot as plt
```

We are creating a DataFrame `df`. Load `colleges.csv` and display the DataFrame below.

This college dataset contains a list of American colleges and their rankings, along with other details such as region, college type, student-to-faculty ratio, etc. In the sections below, you will develop hypotheses, test them, and draw conclusions.

```python
# Your code here
df = pd.read_csv("colleges.csv")
df
```

```
                                           description  rank  \
0    A leading global research university, MIT attr...     1
1    Stanford University sits just outside of Palo ...     2
2    One of the top public universities in the coun...     2
3    Princeton is a leading private research univer...     4
4    Located in upper Manhattan, Columbia Universit...     5
..                                                 ...   ...
493  St. Joseph’s College is a private institution ...   494
494  A liberal arts college founded by the Moravian...   495
495  Lawrence Technological University in Southfiel...   496
496  Saint Martin’s University in Lacey, WA, one of...   497
497  The University of Memphis is a large public re...   498

                          organizationName state  studentPopulation  \
0    Massachusetts Institute of Technology    MA              12195
1                      Stanford University    CA              20961
2         University of California, Berkeley    CA              45878
3                     Princeton University    NJ               8532
4                      Columbia University    NY              33882
..                                     ...   ...                ...
493               St. Joseph's College (NY)    NY               5901
494                     Moravian University    PA               2961
495       Lawrence Technological University    MI               3163
496               Saint Martin's University    WA               1980
497                  University of Memphis    TN              25128

    campusSetting  medianBaseSalary   longitude   latitude  \
```

```
0           Urban          173700.0  -71.093539  42.359006
1        Suburban          173500.0 -122.168924  37.431370
2           Urban          154500.0 -122.258393  37.869236
3           Urban          167600.0  -74.659119  40.349855
4           Urban          148800.0  -73.961288  40.806515
..            ...               ...         ...        ...
493         Urban          100900.0  -73.968304  40.690548
494         Urban          109800.0  -75.381596  40.630303
495         Urban          119900.0  -83.278458  42.450606
496         Urban          102100.0         NaN        NaN
497         Urban           90700.0  -89.939618  35.118453

                       website  ... yearFounded stateCode  \
0           http://web.mit.edu  ...      1861.0        MA
1      http://www.stanford.edu  ...      1891.0        CA
2      http://www.berkeley.edu  ...      1868.0        CA
3     http://www.princeton.edu  ...      1746.0        NJ
4      http://www.columbia.edu  ...      1754.0        NY
..                         ...  ...         ...       ...
493       http://www.sjcny.edu  ...      1916.0        NY
494    http://www.moravian.edu  ...      1742.0        PA
495  http://https://www.ltu.edu  ...         NaN        MI
496                        NaN  ...         NaN        WA
497      http://www.mephis.edu  ...      1912.0        TN

                collegeType  \
0    Private not-for-profit
1    Private not-for-profit
2                    Public
3    Private not-for-profit
4    Private not-for-profit
..                      ...
493  Private not-for-profit
494  Private not-for-profit
495  Private not-for-profit
496  Private not-for-profit
497                  Public

                                carnegieClassification
studentFacultyRatio  \
0    Doctoral Universities: Very High Research Acti...
3
1    Doctoral Universities: Very High Research Acti...
4
2    Doctoral Universities: Very High Research Acti...
19
3    Doctoral Universities: Very High Research Acti...
4
4    Doctoral Universities: Very High Research Acti...
6
```

```
..                                                      ...
...
493  Master's Colleges & Universities: Medium Programs
12
494      Baccalaureate Colleges: Arts & Sciences Focus
11
495  Master's Colleges & Universities: Larger Programs
11
496  Master's Colleges & Universities: Medium Programs
12
497      Doctoral Universities: High Research Activity
16

     totalStudentPop undergradPop totalGrantAid
percentOfStudentsFinAid  \
0              12195         4582    35299332.0
75.0
1              20961         8464    51328461.0
70.0
2              45878        33208    64495611.0
63.0
3               8532         5516    44871096.0
62.0
4              33882         8689    44615007.0
58.0
..               ...          ...           ...                       ..
.
493             5901         4429    11919881.0
99.0
494             2961         2268    12685943.0
100.0
495             3163         2286     5639254.0
97.0
496             1980         1644     9759540.0
99.0
497            25128        20011    27575189.0
98.0

     percentOfStudentsGrant
0                      60.0
1                      55.0
2                      53.0
3                      61.0
4                      54.0
..                      ...
493                    99.0
494                   100.0
495                    96.0
496                    99.0
```

```
497                        97.0

[498 rows x 25 columns]
```

TASK 2.1 (2 POINTS): Some entries of the dataframe are NaN. remove those entries.

```
# Your code here
df = df.dropna()
df
```

```
                                           description  rank  \
0    A leading global research university, MIT attr...     1
1    Stanford University sits just outside of Palo ...     2
2    One of the top public universities in the coun...     2
3    Princeton is a leading private research univer...     4
4    Located in upper Manhattan, Columbia Universit...     5
..                                                 ...   ...
490  Loyola University New Orleans provides student...   491
491  Xavier University is a Jesuit Catholic school ...   492
493  St. Joseph’s College is a private institution ...   494
494  A liberal arts college founded by the Moravian...   495
497  The University of Memphis is a large public re...   498

                          organizationName state  studentPopulation  \
0    Massachusetts Institute of Technology    MA              12195
1                      Stanford University    CA              20961
2       University of California, Berkeley    CA              45878
3                     Princeton University    NJ               8532
4                      Columbia University    NY              33882
..                                     ...   ...                ...
490          Loyola University New Orleans    LA               4972
491                      Xavier University    OH               8079
493              St. Joseph's College (NY)    NY               5901
494                    Moravian University    PA               2961
497                  University of Memphis    TN              25128

    campusSetting  medianBaseSalary   longitude   latitude  \
0           Urban          173700.0  -71.093539  42.359006
1        Suburban          173500.0 -122.168924  37.431370
2           Urban          154500.0 -122.258393  37.869236
3           Urban          167600.0  -74.659119  40.349855
4           Urban          148800.0  -73.961288  40.806515
..            ...               ...         ...        ...
490         Urban          102300.0  -90.077714  29.953690
491         Urban          104900.0  -84.476379  39.149037
493         Urban          100900.0  -73.968304  40.690548
494         Urban          109800.0  -75.381596  40.630303
497         Urban           90700.0  -89.939618  35.118453

                      website  ... yearFounded stateCode  \
```

```
0          http://web.mit.edu  ...      1861.0        MA
1     http://www.stanford.edu  ...      1891.0        CA
2     http://www.berkeley.edu  ...      1868.0        CA
3    http://www.princeton.edu  ...      1746.0        NJ
4     http://www.columbia.edu  ...      1754.0        NY
..                        ...  ...         ...       ...
490      http://www.loyno.edu  ...      1904.0        LA
491     http://www.xavier.edu  ...      1831.0        OH
493      http://www.sjcny.edu  ...      1916.0        NY
494   http://www.moravian.edu  ...      1742.0        PA
497     http://www.mephis.edu  ...      1912.0        TN

                collegeType  \
0    Private not-for-profit
1    Private not-for-profit
2                    Public
3    Private not-for-profit
4    Private not-for-profit
..                      ...
490  Private not-for-profit
491  Private not-for-profit
493  Private not-for-profit
494  Private not-for-profit
497                  Public

                               carnegieClassification
studentFacultyRatio  \
0    Doctoral Universities: Very High Research Acti...
3
1    Doctoral Universities: Very High Research Acti...
4
2    Doctoral Universities: Very High Research Acti...
19
3    Doctoral Universities: Very High Research Acti...
4
4    Doctoral Universities: Very High Research Acti...
6
..                                                ...
...
490                 Doctoral/Professional Universities
13
491  Master's Colleges & Universities: Larger Programs
11
493  Master's Colleges & Universities: Medium Programs
12
494      Baccalaureate Colleges: Arts & Sciences Focus
11
497      Doctoral Universities: High Research Activity
16
```

```
     totalStudentPop undergradPop totalGrantAid  
percentOfStudentsFinAid  \
0              12195         4582    35299332.0  
75.0
1              20961         8464    51328461.0  
70.0
2              45878        33208    64495611.0  
63.0
3               8532         5516    44871096.0  
62.0
4              33882         8689    44615007.0  
58.0
..               ...          ...           ...                      ..
.
490             4972         3538    26114959.0  
99.0
491             8079         5473    28294277.0  
100.0
493             5901         4429    11919881.0  
99.0
494             2961         2268    12685943.0  
100.0
497            25128        20011    27575189.0  
98.0

     percentOfStudentsGrant  
0                      60.0  
1                      55.0  
2                      53.0  
3                      61.0  
4                      54.0  
..                      ...  
490                    99.0  
491                   100.0  
493                    99.0  
494                   100.0  
497                    97.0  

[422 rows x 25 columns]
```

#Q3) (8 POINTS) Hypothesis Testing

Try to find relationships in this dataset through hypothesis testing. For each hypothesis test:

- First chose a null hypothesis, or a statement that there is no effect between different variables, that serves as a default assumption.
- Then chose an alternative hypothesis, or a statement that suggests that there is a correlation between different variables.

For the questions below, assume $\alpha = 0.05$.

First Hypothesis

- HO: The region of the college does not have an effect on the likelihood of the college type.
- HA: The region of the college does have an effect on the likelihood of the college type.

Our plan is to apply a chi-squared test. You may find it helpful to consult the `scipy.stats` library's documentation: https://docs.scipy.org/doc/scipy/reference/stats.html

Contingency table is a table used in statistics to display the frequency distribution of variables. It will help us perform a chi-squared test on our data. You can find more information on contingency table here - https://en.wikipedia.org/wiki/Contingency_table

TASK 3.1 (2 POINTS): Create a contingency table and display it.

```
# Your code here
contingency_table = pd.crosstab(df['region'], df['collegeType'])
contingency_table
```

collegeType	Private not-for-profit	Public
region		
Midwest	54	37
Northeast	109	39
South	37	57
West	37	52

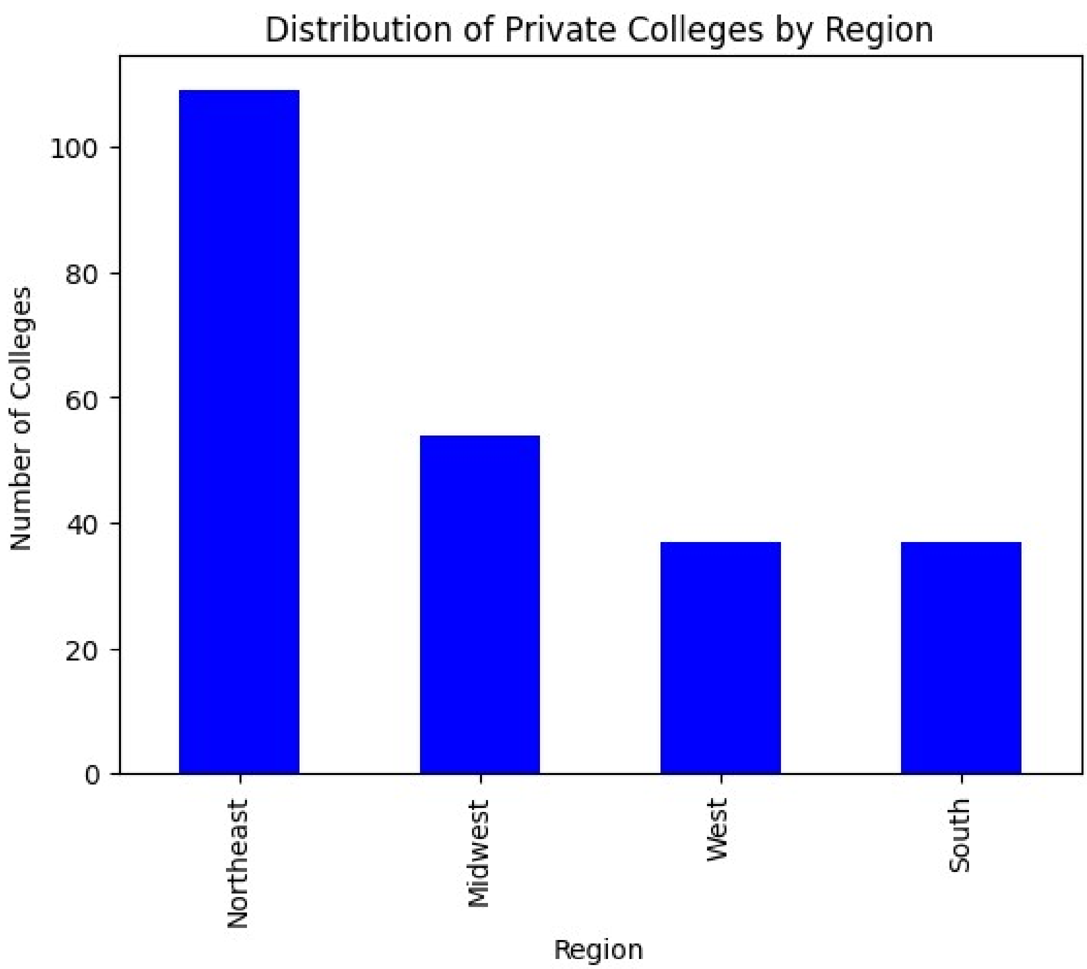
TASK 3.2 (2 POINTS): Why would we consider using a chi-squared test specifically (as opposed to some other hypothesis test)?

> A chi-squared test is the right choice because the variables are categories like regions and college types instead of numerical measurements. This test specifically checks if there is a real connection between these groups or if the distribution is just random. Since calculating an average for a "region" is impossible, a test that compares frequencies in a contingency table is necessary. Using this method helps determine if the location of a school actually relates to whether it is public or private.

TASK 3.3 (2 POINTS): Create a plot showing the relationship between the regions and the no. of private colleges in it.

```
# Any graph that can be easily identified as the prompt above is
suffice
private_colleges = df[df['collegeType'].str.contains('Private')]

# count occurrences of private schools in each region
private_counts = private_colleges['region'].value_counts()

private_counts.plot(kind='bar', color='blue')
```

```python
plt.title('Distribution of Private Colleges by Region')
plt.ylabel('Number of Colleges')
plt.xlabel('Region')
plt.show()
```



TASK 3.4 (2 POINTS): Explain what you can infer from your plot

> From the plot, it looks like the Northeast has the most private colleges by a pretty large margin compared to the other regions. The Midwest has a moderate amount, while the West and South both have noticeably fewer and seem fairly close to each other. Overall, it seems like private colleges are much more concentrated in the Northeast than in the rest of the country.

#Q4) (5 POINTS) Conduct the chi-squared test

TASK: 4.1 (2 POINTS): Display the p-value of applying the chi-squared test using the `chi2_contingency()` function.

```python
# Your code here
from scipy.stats import chi2_contingency
```

```
chi2, p_value, dof, expected = chi2_contingency(contingency_table)

p_value

4.123685954796126e-08
```

TASK: 4.2 (3 POINTS): Based on the p-value, determine whether to reject or fail to reject the null hypothesis. Explain your answer.

> The p-value from the chi-square test is approximately 4.12×10^(-8), which is much smaller than the significance level of 0.05. Because of this, we reject the null hypothesis. This suggests that there is a relationship between region and college type, meaning the distribution of college types is not independent of the region.

#Q5) (3 POINTS) A New Hypothesis

Now create a new hypothesis test for whether the campus setting has an effect on the total student population. (Assume $\alpha = 0.05$).

TASK 5.1 (3 POINTS): Write down your null and alternative hypotheses:

> Null hypothesis: Campus setting has no effect on the total student population, meaning the average student population is the same across different campus settings.

Alternative hypothesis: Campus setting does have an effect on the total student population, meaning the average student population differs between at least some campus settings.

#Q6) (7 POINTS) Hypothesis Testing

TASK 6.0: Split the data into 3 different dataframes based on campus setting.

```
# YOUR CODE HERE
# split full dataframe into separate dataframes by campus setting
settings = df["campusSetting"].dropna().unique()

# build dictionary: setting name -> subset dataframe
dfs_by_setting = {s: df[df["campusSetting"] == s].copy() for s in
settings}

setting_names = list(dfs_by_setting.keys())
df_setting_1 = dfs_by_setting[setting_names[0]]
df_setting_2 = dfs_by_setting[setting_names[1]]
df_setting_3 = dfs_by_setting[setting_names[2]]

setting_names

['Urban', 'Suburban', 'Rural']
```

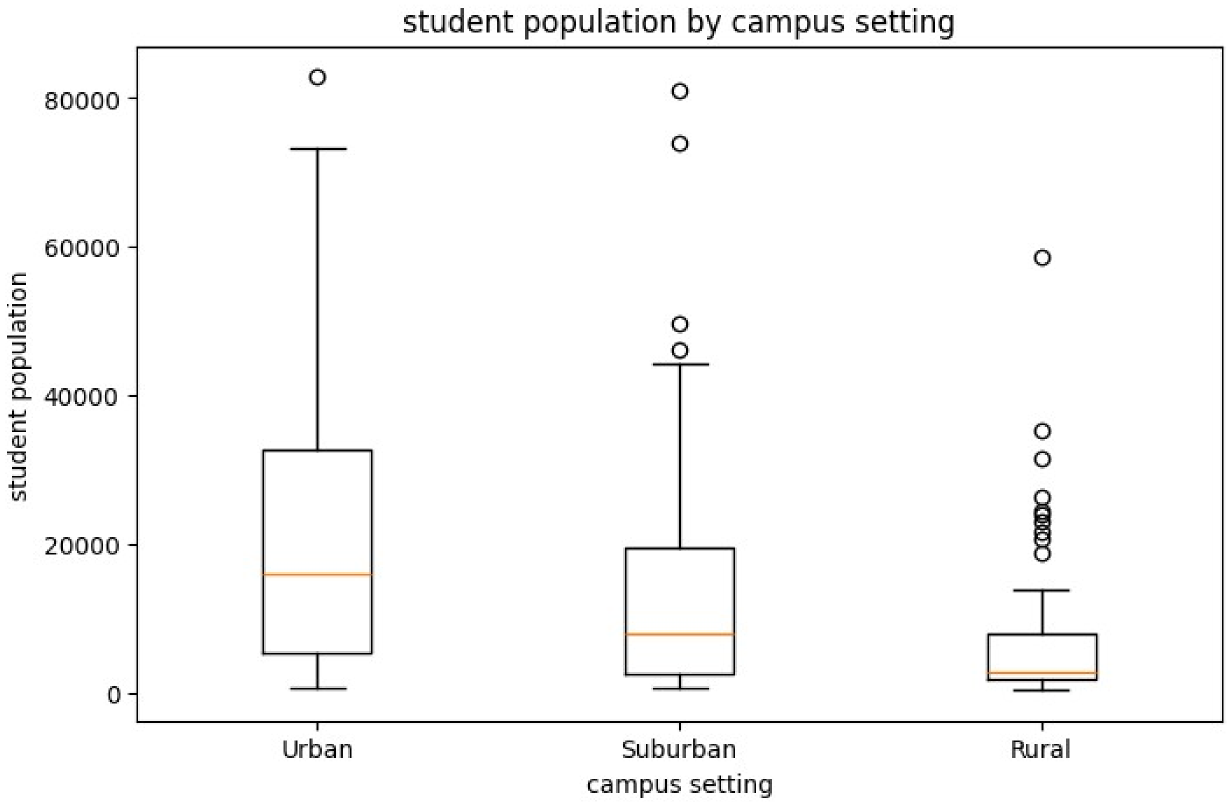
TASK 6.1 (2 POINTS): Choose an appropriate hypothesis test and display the p-value of applying the that test.

```python
# YOUR CODE HERE
from scipy.stats import f_oneway

# extract student populations for each campus setting
groups = [
    dfs_by_setting[s]["studentPopulation"].dropna().astype(float)
    for s in setting_names
]

# run one-way anova (tests whether at least one group mean differs)
f_stat, p_value = f_oneway(*groups)

p_value
```

```
5.371995673404977e-09
```

TASK 6.2 (2 POINTS): Create a graph(s) using `matplotlib` to show the relationship between campus setting and total student population based on the "studentPopulation" variable.

```python
# YOUR CODE HERE
import matplotlib.pyplot as plt

# boxplot to compare student population distributions by campus
setting
plt.figure(figsize=(8, 5))
plt.boxplot(groups, tick_labels=setting_names)

plt.title("student population by campus setting")
plt.xlabel("campus setting")
plt.ylabel("student population")

plt.show()
```



TASK 6.3 (3 POINTS): Based on the p-value, determine whether to reject or fail to reject the null hypothesis. Explain your answer.

> Because the anova p-value is about 5.37×10^{-9}, which is way smaller than $\alpha = 0.05$, we reject the null hypothesis. This means there is strong evidence that campus setting (urban/suburban/rural) affects total student population. The boxplot also supports this since the distributions (especially the medians) look noticeably different across the three settings.

#*Q7)* (2 POINTS) Post Hoc Tests

TASK 7.1 (2 POINTS): Why might we need post-hoc tests in this scenario?

> Anova only tells us that at least one group mean is different, but it doesn't tell us which specific campus settings differ from each other. We need post-hoc tests to compare the groups pairwise (urban vs suburban, urban vs rural, suburban vs rural) and figure out where the differences actually are, while controlling for making multiple comparisons.

BONUS TASK 7.2 (2 POINTS): Apply a post-hoc test of your choice

> Write your interpretation here

#*Q8)* (19 POINTS) Hypothesis Test

Now create a new hypothesis test for whether the total grant aid has an effect on college ranking. (Assume $\alpha = 0.05$).

TASK 8.1 (3 POINTS): Write down the null and alternative hypotheses below.

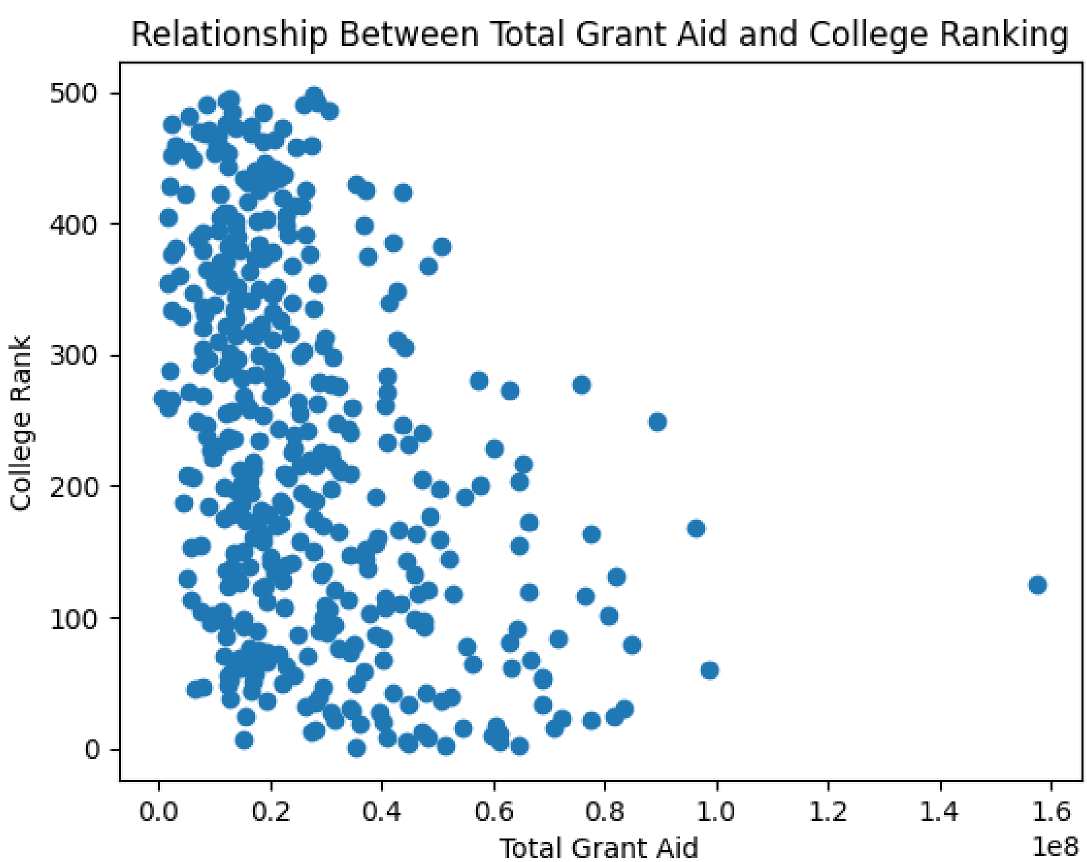
Null hypothesis: There is no relationship between total grant aid and college ranking.

Alternative hypothesis: There is a relationship between total grant aid and college ranking.

TASK 8.2 (2 POINTS): Create a plot using `matplotlib` that visualizes your hypothesis.

Note: Plot the actual data in a way that depicts your hypothesis. You may write a sentence or two underneath the plot explaining how the plot relates to your hypothesis.

```
# YOUR CODE HERE
import matplotlib.pyplot as plt

plt.scatter(df['totalGrantAid'], df['rank'])

plt.xlabel('Total Grant Aid')
plt.ylabel('College Rank')
plt.title('Relationship Between Total Grant Aid and College Ranking')

plt.show()
```



TASK 8.3 (3 POINTS): Apply an appropriate hypothesis test and find the p-value of it. Only for this question, you are allowed to apply a hypothesis test that we haven't covered in the Hypothesis Testing class (hints: how about we consider finding some "relation" between them)?

```
# YOUR CODE HERE
from scipy.stats import pearsonr

# remove missing values
data = df[['totalGrantAid', 'rank']].dropna()

corr, p_value = pearsonr(data['totalGrantAid'], data['rank'])

p_value
```

```
1.6476144412424664e-19
```

TASK 8.4 (3 POINTS): Based on the p-value, determine whether to reject or fail to reject the null hypothesis. Explain your answer.

> Since the p-value is about 1.65×10^{-19}, which is much smaller than 0.05, we reject the null hypothesis. This means there is very strong evidence of a relationship between total grant aid and college ranking. In other words, total grant aid and rank do not appear to be independent in this dataset.

TASK 8.5 (3 POINTS): Based on your previous answer, can you conclude that increasing grant aid will change a college's ranking? What is experimental procedure required to reach this conclusion?

> No, we cannot conclude that increasing grant aid will definitely change a college's ranking just from this test. What we found is a relationship, but correlation does not prove causation. To make a conclusion, we would need some kind of controlled experiment or a very strong quasi-experimental design where changes in grant aid are isolated from other factors that also affect ranking.

TASK 8.6 (3 POINTS): What kind of t-test (right-tail or left-tail) would you use to verify the following hypothesis?

H0: There is no difference in student to faculty ratio between private and public colleges

HA: Private colleges have a smaller student to faculty ratio

Also perform the test and print your p value.

> I would use a left-tailed t-test. This is because the alternative hypothesis says that private colleges have a smaller student-to-faculty ratio than public colleges, meaning we are specifically testing whether the mean for private colleges is less than the mean for public colleges.

```
# your code here
from scipy.stats import ttest_ind

# separate student to faculty ratios
```

```python
private_ratio = df[df["collegeType"] == "Private not-for-profit"]
["studentFacultyRatio"].dropna().astype(float)
public_ratio = df[df["collegeType"] == "Public"]
["studentFacultyRatio"].dropna().astype(float)

# perform two-sample t-test
t_stat, p_two_tail = ttest_ind(private_ratio, public_ratio,
equal_var=False)

# convert to one-tailed (left-tail) p-value
p_value = p_two_tail / 2

p_value
```

```
5.82573591572242e-66
```

TASK 8.7 (2 POINTS): Based on the p-value, determine whether to reject or fail to reject the null hypothesis. Explain your answer.

> The p-value (5.83×10^{-66}) is much smaller than 0.05, so we reject the null hypothesis. This provides strong evidence that private colleges have a smaller student-to-faculty ratio than public colleges.

THE END!